

Introduction to Artificial Intelligence

Language

Natural Language Processing

Natural Language Processing

- automatic summarization
- information extraction
- language identification
- machine translation
- named entity recognition
- speech recognition
- text classification
- word sense disambiguation
- ...

Syntax

"Just before nine o'clock Sherlock Holmes stepped briskly into the room."

"Just before Sherlock Holmes nine o'clock stepped briskly the room."

"I saw the man on the mountain
with a telescope."

Semantics

"Just before nine o'clock Sherlock
Holmes stepped briskly into the room."

"Sherlock Holmes stepped briskly into the room just before nine o'clock."

"A few minutes before nine, Sherlock Holmes walked quickly into the room."

"Colorless green ideas sleep furiously."

Natural Language Processing

Syntax

formal grammar

a system of rules for generating sentences
in a language

Context-Free Grammar

she

saw

the

city

N
|
she

V
|
saw

D
|
the

N
|
city

N → she | city | car | Harry | ...

D → the | a | an | ...

V → saw | ate | walked | ...

P → to | on | over | ...

ADJ → blue | busy | old | ...

$NP \rightarrow N \mid D N$

$NP \rightarrow N \mid DN$

NP

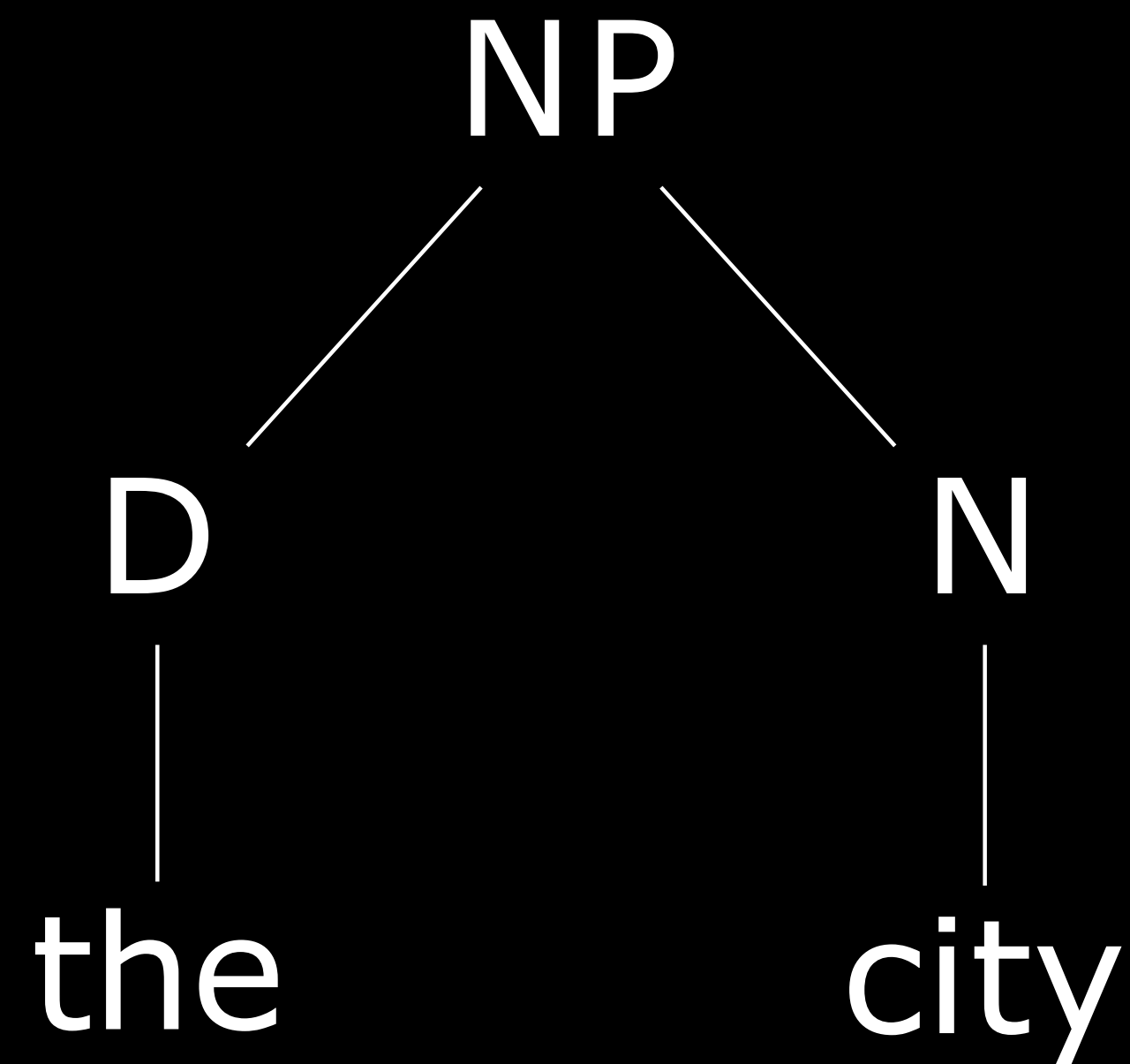


N



she

$NP \rightarrow N \mid D N$

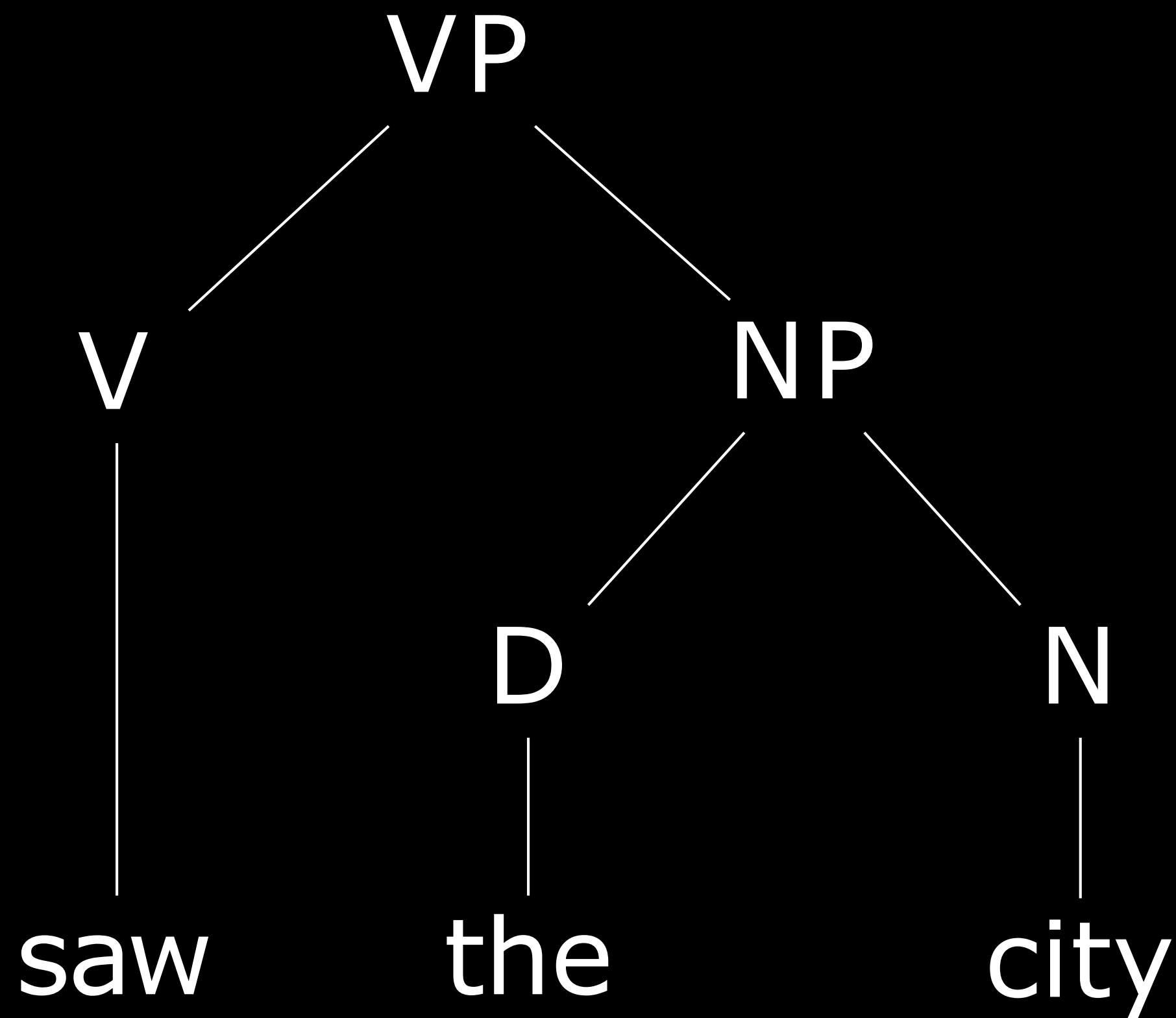


$VP \rightarrow V \mid V NP$

$VP \rightarrow V \mid V NP$

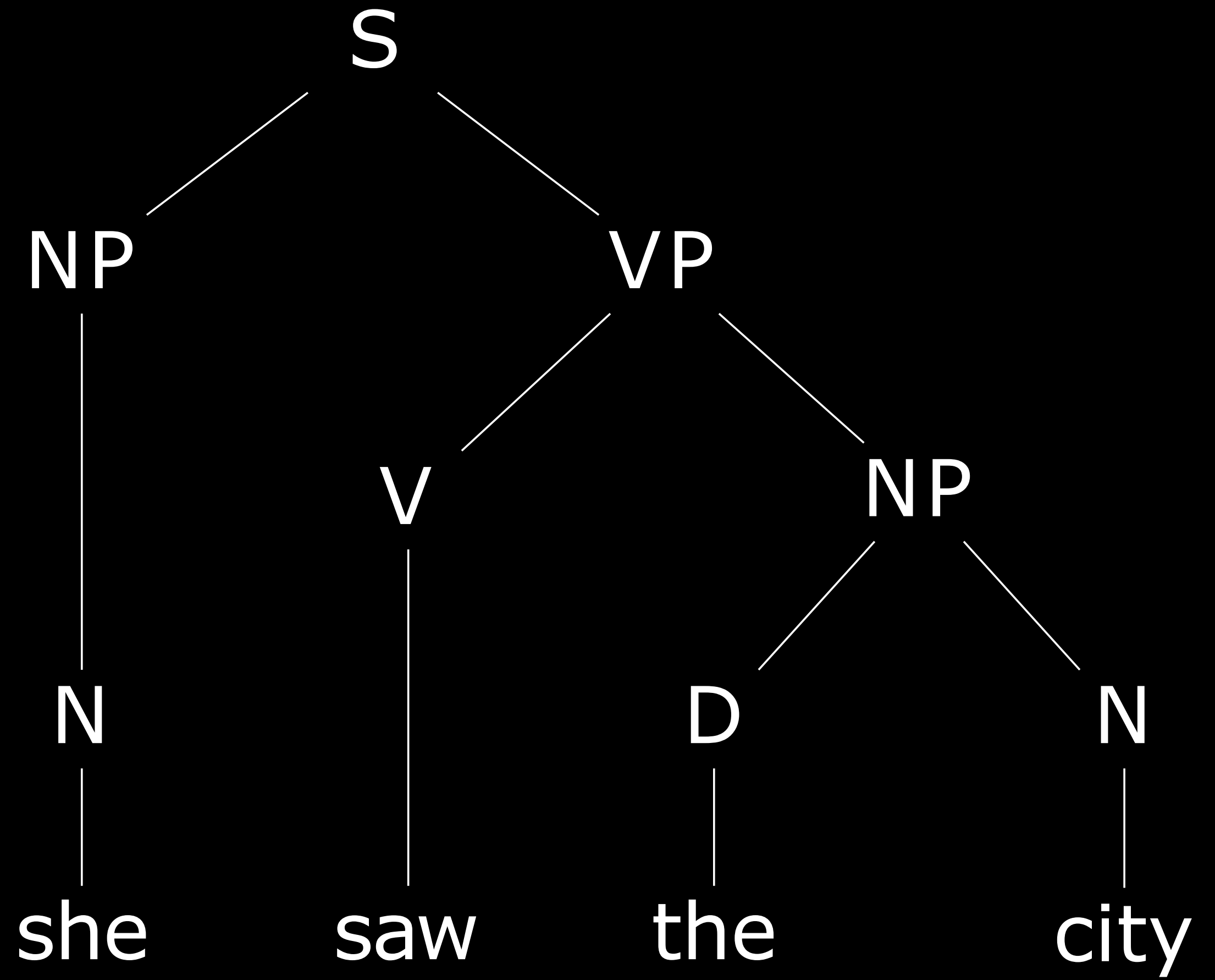
VP
|
V
|
walked

$VP \rightarrow V \mid V NP$



$S \rightarrow NP VP$

$S \rightarrow NP VP$



nlTK

n-gram

a contiguous sequence of *n* items
from a sample of text

character n -gram

a contiguous sequence of n characters
from a sample of text

word n -gram

a contiguous sequence of n words
from a sample of text

unigram

a contiguous sequence of 1 item
from a sample of text

bigram

a contiguous sequence of 2 items
from a sample of text

trigrams

a contiguous sequence of 3 items
from a sample of text

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

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"How often have I **said to you** that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

"How often have I said to you that when you have eliminated the impossible whatever remains, however improbable, must be the truth?"

tokenization

the task of splitting a sequence of
characters into pieces (tokens)

word tokenization

the task of splitting a sequence of
characters into words

sentence tokenization

the task of splitting a sequence of
characters into sentences

"Whatever remains, however improbable, must be the truth."

"Whatever remains, however
improbable, must be the truth."

["Whatever", "remains,", "however",
"improbable,", "must", "be", "the",
"truth."]

"Whatever remains, however
improbable, must be the truth."

["Whatever", "remains,", "however",
"improbable,", "must", "be", "the",
"truth."]

"Whatever remains, however
improbable, must be the truth."

["Whatever", "remains", "however",
"improbable", "must", "be", "the",
"truth"]

"Just before nine o'clock Sherlock Holmes stepped briskly into the room."

"Just before nine o'clock Sherlock
Holmes stepped briskly into the room."

"He was dressed in a sombre yet rich style, in black frock-coat, shining hat, neat brown gaiters, and well-cut pearl-grey trousers."

"He was dressed in a sombre yet rich style, in black frock-coat, shining hat, neat brown gaiters, and well-cut pearl-grey trousers."

"I cannot waste time over this sort of fantastic talk, Sherlock. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Sherlock. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Sherlock. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Sherlock. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Mr. Holmes. If you can catch the man, catch him, and let me know when you have done it."

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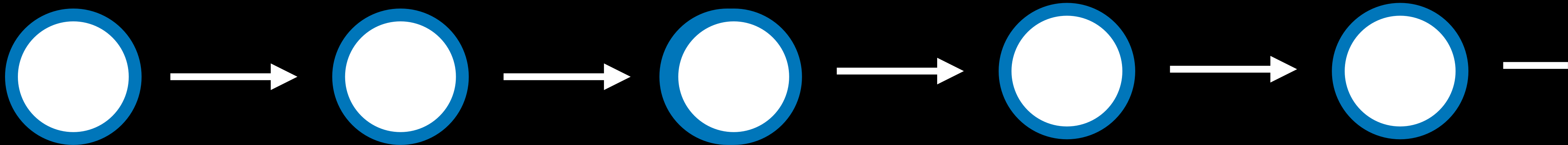
"I cannot waste time over this sort of fantastic talk, Mr. **Holmes**. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Mr. Holmes. If you can catch the man, catch him, and let me know when you have done it."

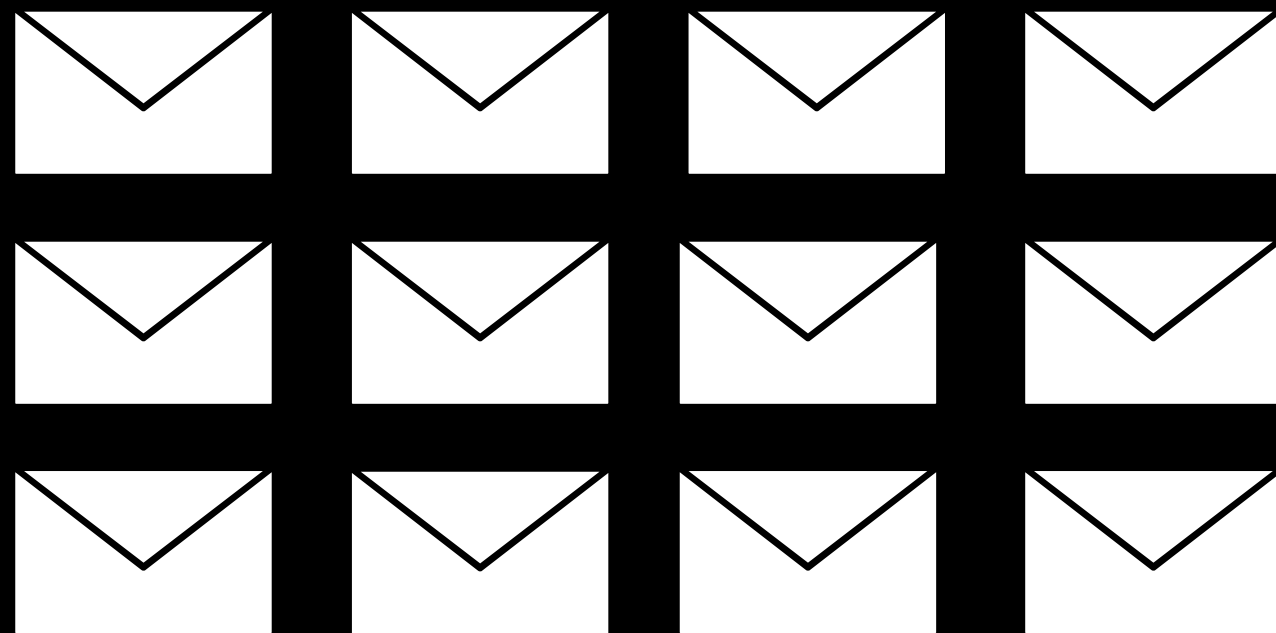
"I cannot waste time over this sort of fantastic talk, Mr. Holmes. If you can catch the man, catch him, and let me know when you have done it."

"I cannot waste time over this sort of fantastic talk, Mr. Holmes," he said. "If you can catch the man, catch him, and let me know when you have done it."

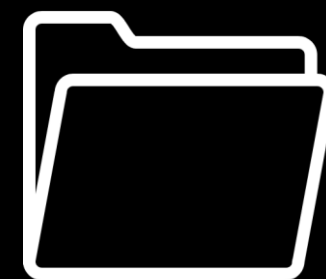
Markov Models



Text Categorization



Inbox



Spam



"My grandson loved it! So much fun!"

"Product broke after a few days."

"One of the best games I've played in a long time."

"Kind of cheap and flimsy, not worth it."



"My grandson loved it! So much fun!"



"Product broke after a few days."



"One of the best games I've played in a long time."



"Kind of cheap and flimsy, not worth it."



"My grandson loved it! So much fun!"



"Product broke after a few days."



"One of the best games I've played in a long time."



"Kind of cheap and flimsy, not worth it."

bag-of-words model

model that represents text as an unordered
collection of words

Naive Bayes

Bayes' Rule

$$P(b \mid a) = \frac{P(a \mid b) P(b)}{P(a)}$$

$P(\text{Positive})$

$P(\text{Negative})$

$P(\text{😊})$

$P(\text{😞})$

"My grandson loved it!"

$$P(\text{😊})$$

$P(\text{😊} \mid \text{"my grandson loved it"})$

$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$

$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

equal to

$$P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"} \mid \text{😊}) P(\text{😊})$$

$$P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

proportional to

$$P(\text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"} \mid \text{😊}) P(\text{😊})$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

proportional to

$$P(\text{😊}, \text{" my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

$$P(\text{😊} \mid \text{"my"}, \text{"grandson"}, \text{"loved"}, \text{"it"})$$

naively proportional to

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

$$P(\text{😊}) = \frac{\text{number of positive samples}}{\text{number of total samples}}$$

$$P(\text{"loved"} \mid \text{😊}) = \frac{\text{number of positive samples with "loved"}}{\text{number of positive samples}}$$

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

😊	😞
0.49	0.51

	😊	😞
my	0.30	0.20
grandson	0.01	0.02
loved	0.32	0.08
it	0.30	0.40

$$P(\text{😊})P(\text{"my"} \mid \text{😊})P(\text{"grandson"} \mid \text{😊}) \\ P(\text{"loved"} \mid \text{😊}) P(\text{"it"} \mid \text{😊})$$

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0.49	0.51

😊
0.00014112

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😄	😞
0.49	0.51

😄 0.00014112

😞 0.00006528

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loved	0.32	0.08
it	0.30	0.40

$$P(\text{😞})P(\text{"my"} \mid \text{😞})P(\text{"grandson"} \mid \text{😞}) \\ P(\text{"loved"} \mid \text{😞}) P(\text{"it"} \mid \text{😞})$$

😄	😞
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😄 0.00014112

😞 0.00006528

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😊	😞
0.49	0.51

😊 0.6837

😞 0.3163

	😊	😞
my	0.30	0.20
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$$P(\text{😞})P(\text{"my"} \mid \text{😞})P(\text{"grandson"} \mid \text{😞}) \\ P(\text{"loved"} \mid \text{😞}) P(\text{"it"} \mid \text{😞})$$

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😄	😞
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😄 0.00000000

😞 0.00006528

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it	0.30	0.40

additive smoothing

adding a value α to each value in our distribution to smooth the data

Laplace smoothing

adding 1 to each value in our distribution:
pretending we've seen each value one more
time than we actually have

information retrieval

the task of finding relevant documents in
response to a user query

topic modeling

models for discovering the topics for a set of documents

term frequency

number of times a term appears in a document

function words

words that have little meaning on their own,
but are used to grammatically connect
other words

function words

am, by, do, is, which, with, yet, ...

content words

words that carry meaning independently

content words

algorithm, category, computer, ...

inverse document frequency

measure of how common or rare a word is
across documents

inverse document frequency

$$\log \frac{\textit{TotalDocuments}}{\textit{NumDocumentsContaining}(\textit{word})}$$

tf-idf

ranking of what words are important in a document by multiplying term frequency (TF) by inverse document frequency (IDF)

Semantics

information extraction

the task of extracting knowledge from
documents

"When Facebook was founded in 2004, it began with a seemingly innocuous mission: to connect friends. Some seven years and 800 million users later, the social network has taken over most aspects of our personal and professional lives, and is fast becoming the dominant communication platform of the future."

Harvard Business Review, 2011

"Remember, back when Amazon was founded in 1994, most people thought his idea to sell books over this thing called the internet was crazy. A lot of people had never even heard of the internet."

Business Insider, 2018

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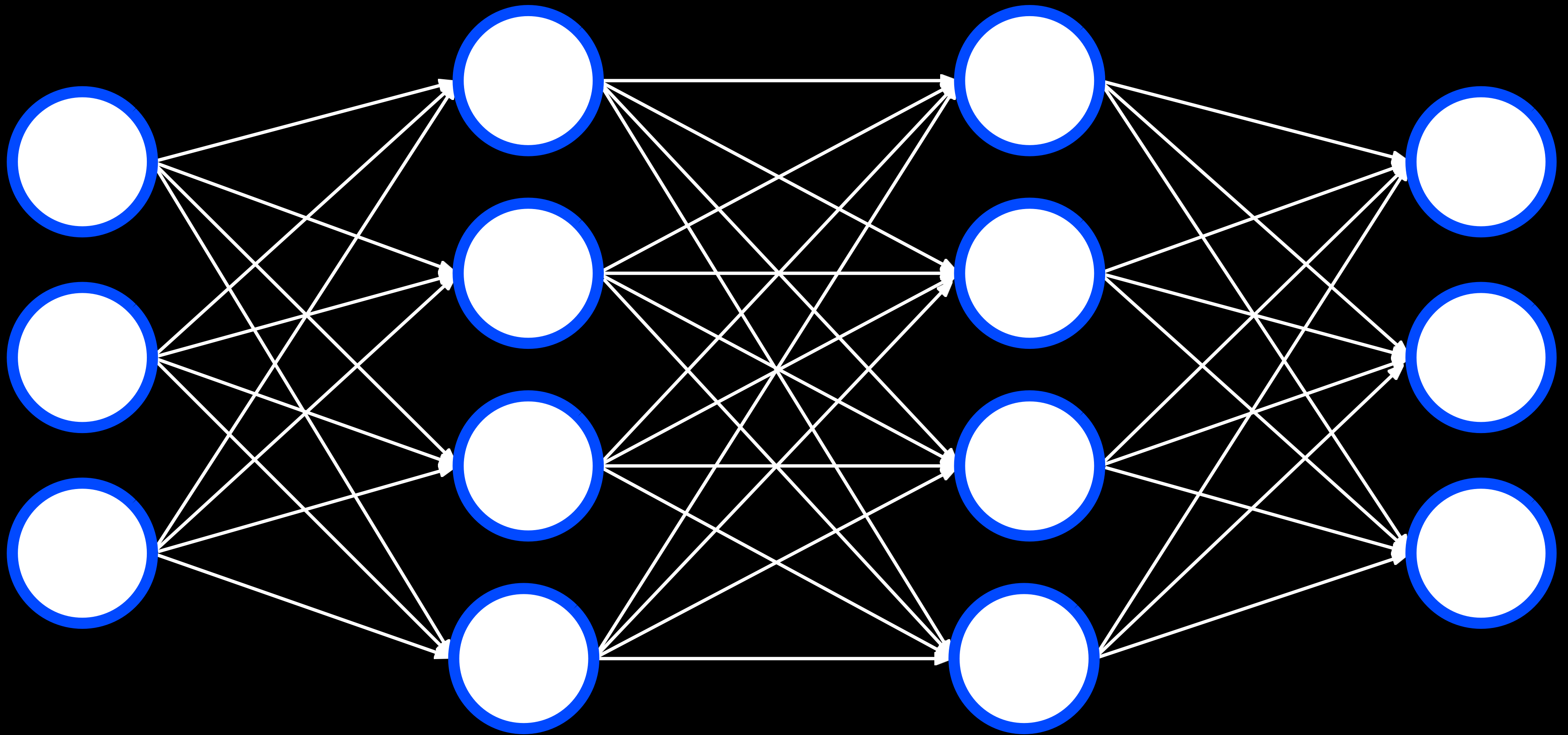
"Remember, back when Amazon was founded in 1994, most people thought his idea to sell books over this thing called the internet was crazy. A lot of people had never even heard of the internet."

Business Insider, 2018

When {company} was founded in {year},

WordNet

Word Representation



"He wrote a book."

he	[1, 0, 0, 0]
----	--------------

wrote	[0, 1, 0, 0]
-------	--------------

a	[0, 0, 1, 0]
---	--------------

book	[0, 0, 0, 1]
------	--------------

one-hot representation

representation of meaning as a vector with a single 1, and with other values as 0

"He wrote a book."

he	[1, 0, 0, 0]
----	--------------

wrote	[0, 1, 0, 0]
-------	--------------

a	[0, 0, 1, 0]
---	--------------

book	[0, 0, 0, 1]
------	--------------

"He wrote a book."

he [1, 0, 0, 0, 0, 0, 0, 0, ...]

wrote [0, 1, 0, 0, 0, 0, 0, 0, ...]

a [0, 0, 1, 0, 0, 0, 0, 0, ...]

book [0, 0, 0, 1, 0, 0, 0, 0, ...]

"He wrote a book."

"He authored a novel."

wrote [0, 1, 0, 0, 0, 0, 0, 0, 0, 0]

authored [0, 0, 0, 0, 1, 0, 0, 0, 0, 0]

book [0, 0, 0, 0, 0, 0, 1, 0, 0, 0]

novel [0, 0, 0, 0, 0, 0, 0, 0, 0, 1]

distribution representation

representation of meaning distributed
across multiple values

"He wrote a book."

he [-0.34, -0.08, 0.02, -0.18, 0.22, ...]

wrote [-0.27, 0.40, 0.00, -0.65, -0.15, ...]

a [-0.12, -0.25, 0.29, -0.09, 0.40, ...]

book [-0.23, -0.16, -0.05, -0.57, ...]

"You shall know a word
by the company it keeps."

J. R. Firth, 1957

for		he	ate
-----	--	----	-----

for	breakfast	he	ate
-----	-----------	----	-----

for

lunch

he

ate

for	dinner	he	ate
-----	--------	----	-----

for		he	ate
-----	--	----	-----

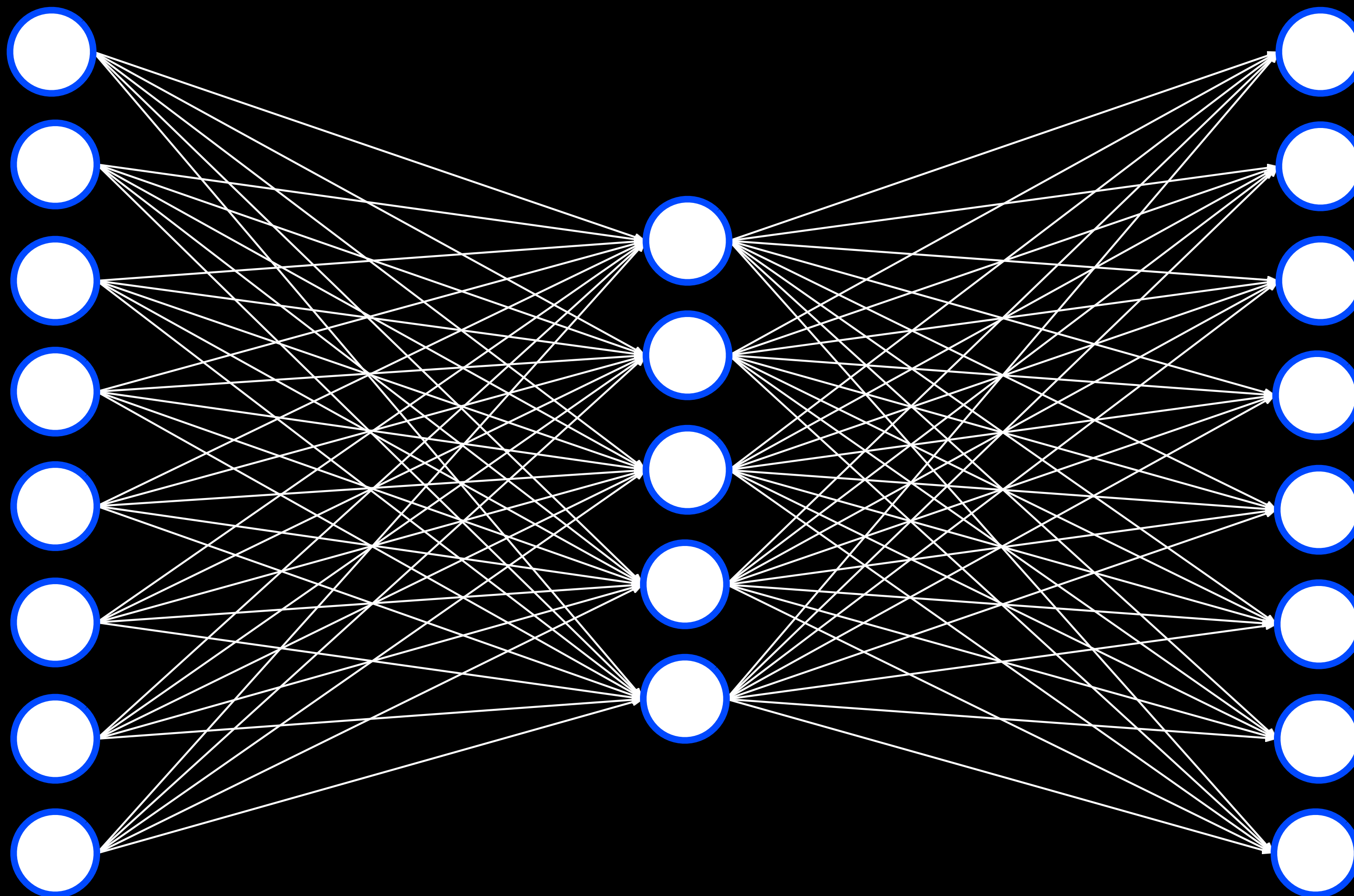
word2vec

model for generating word vectors

skip-gram architecture

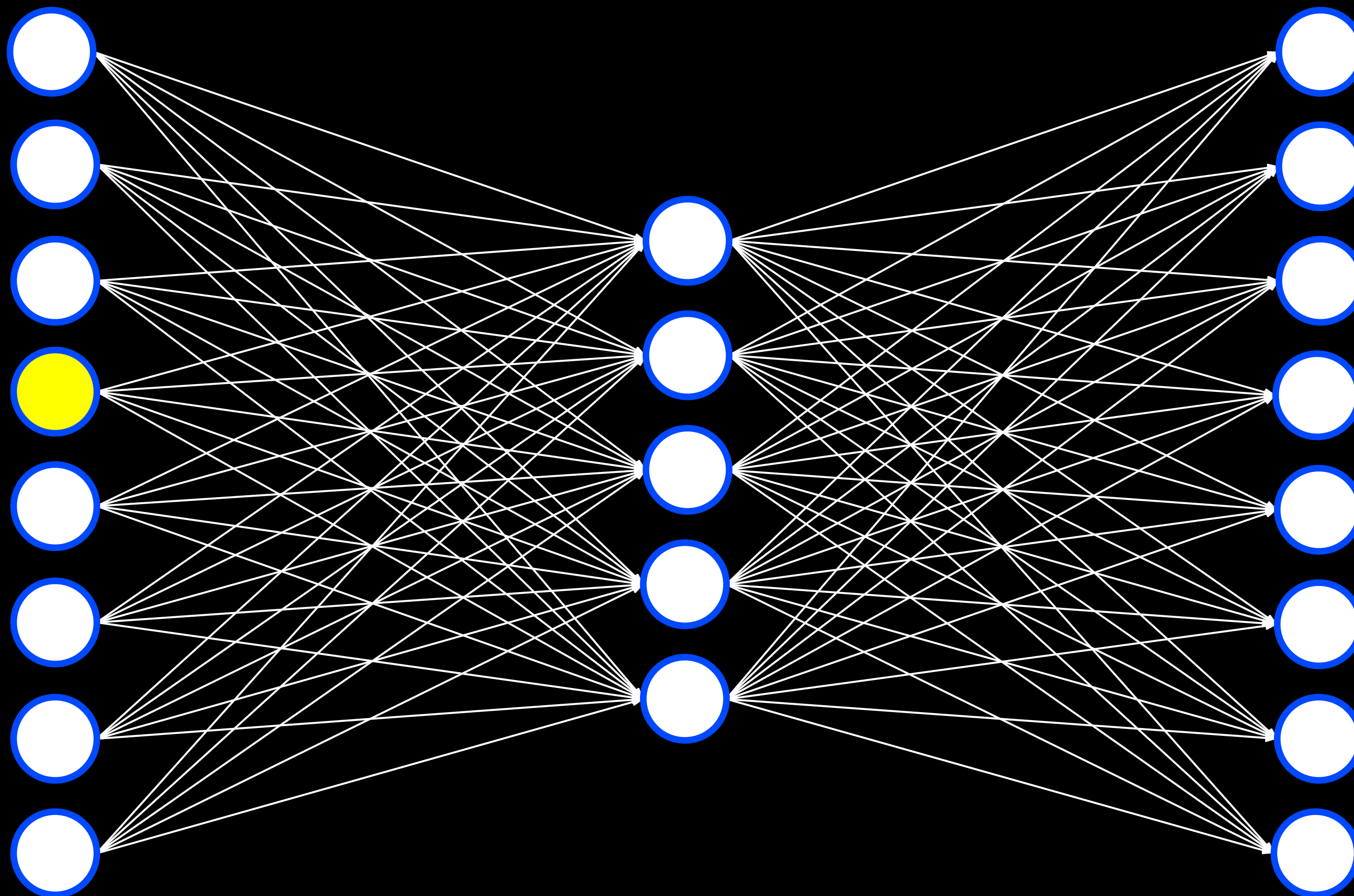
neural network architecture for predicting
context words given a target word

target
word



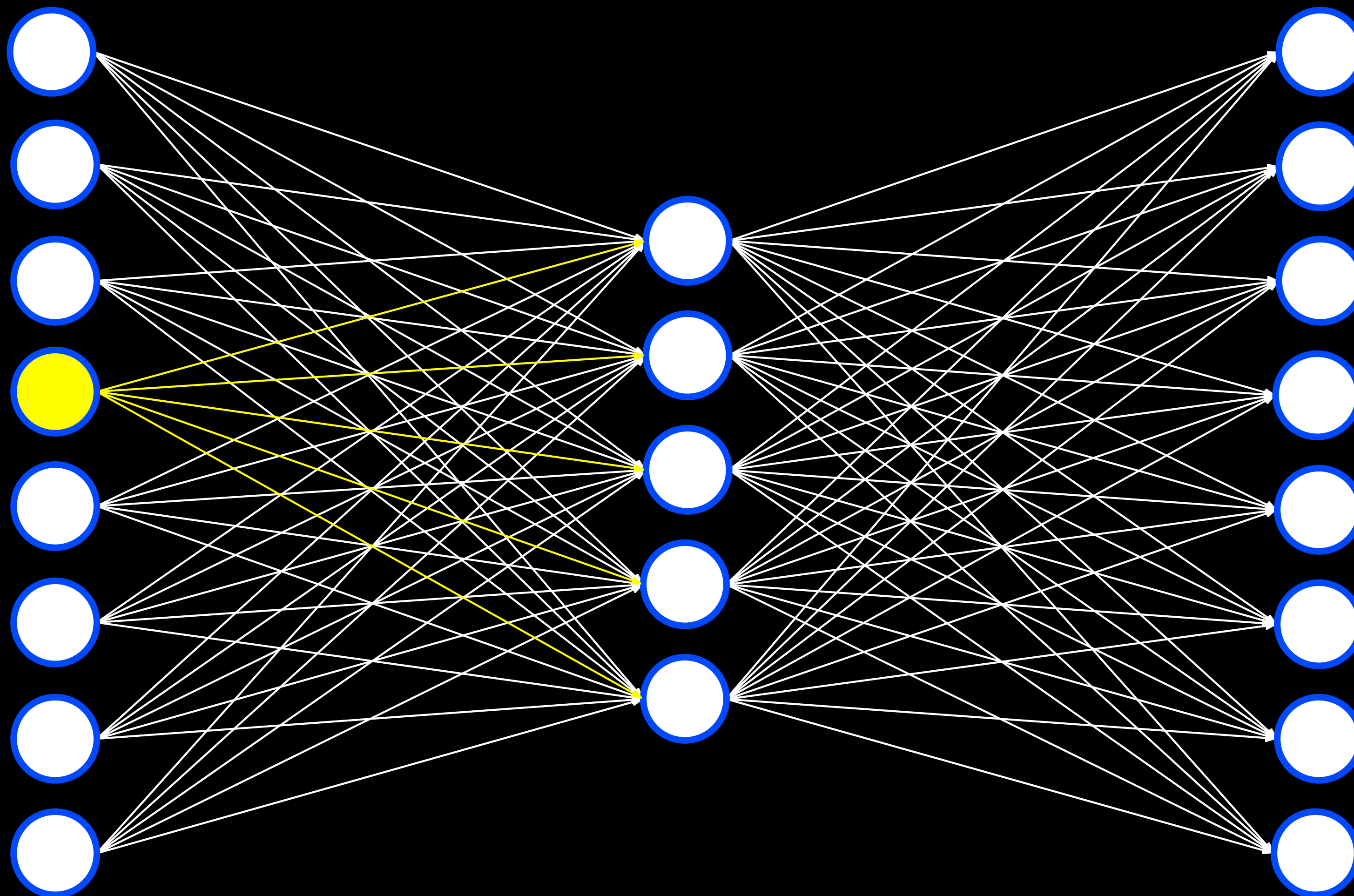
context
word

target
word



context
word

target
word



context
word

breakfast

book

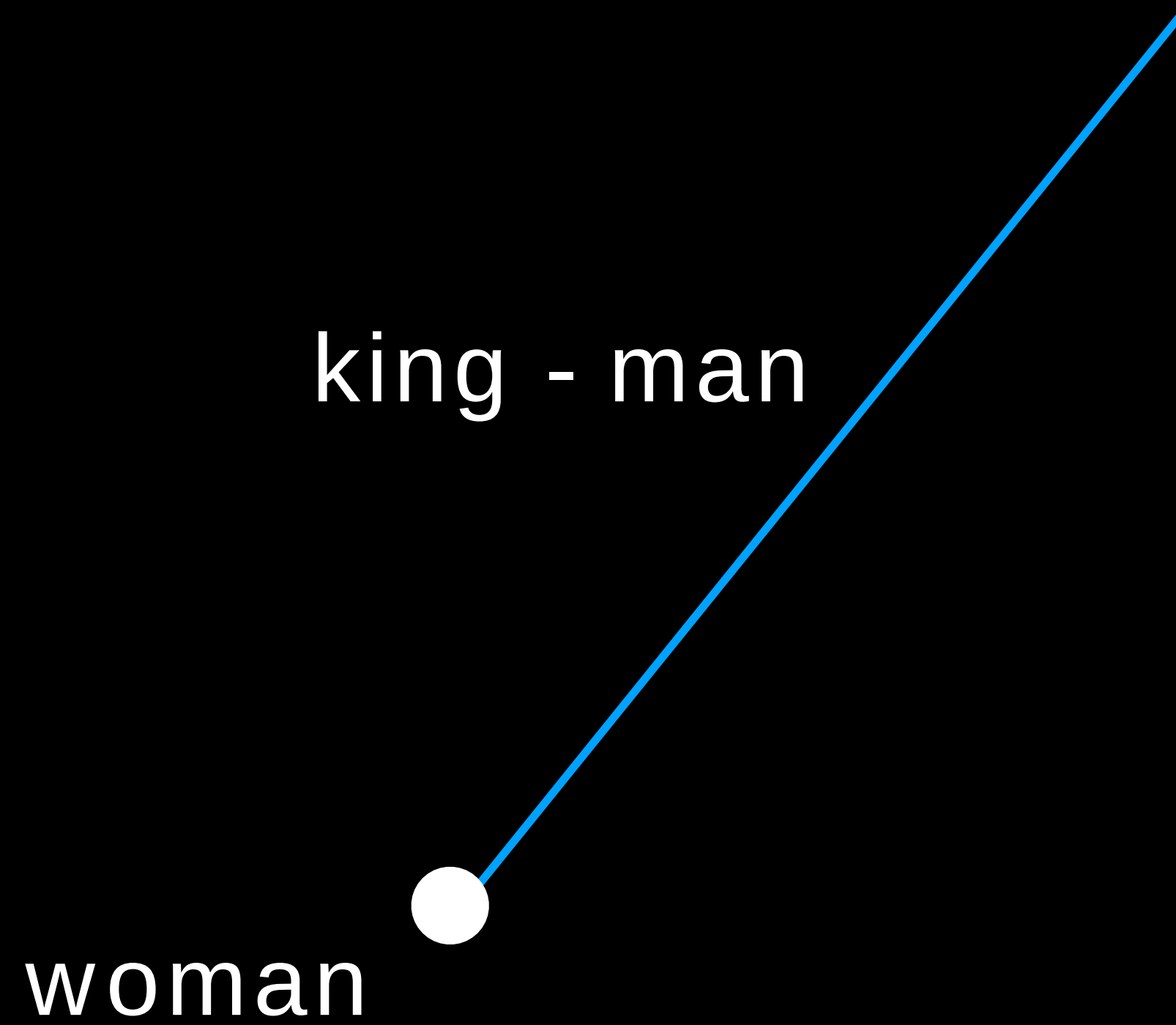
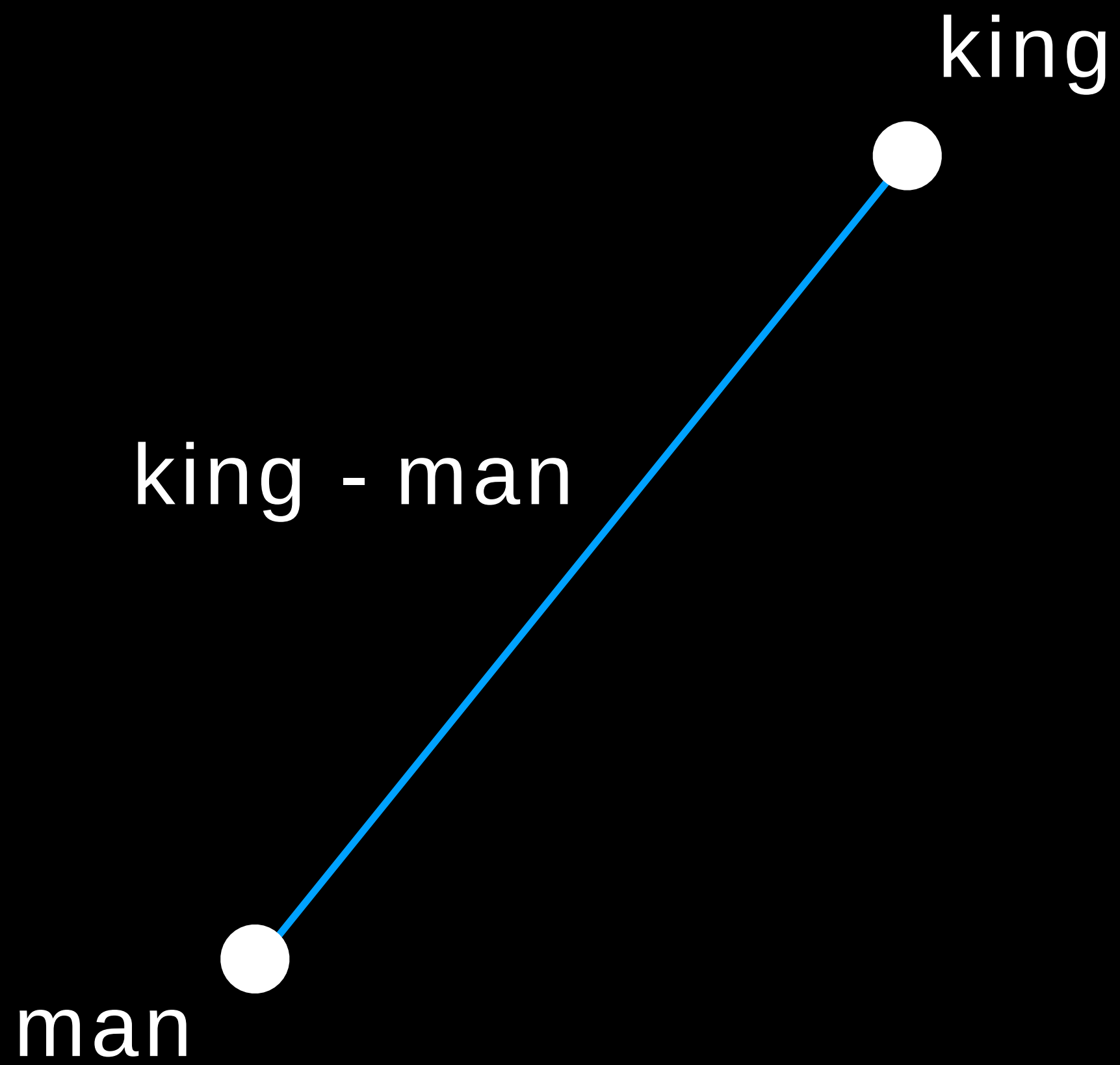
memoir

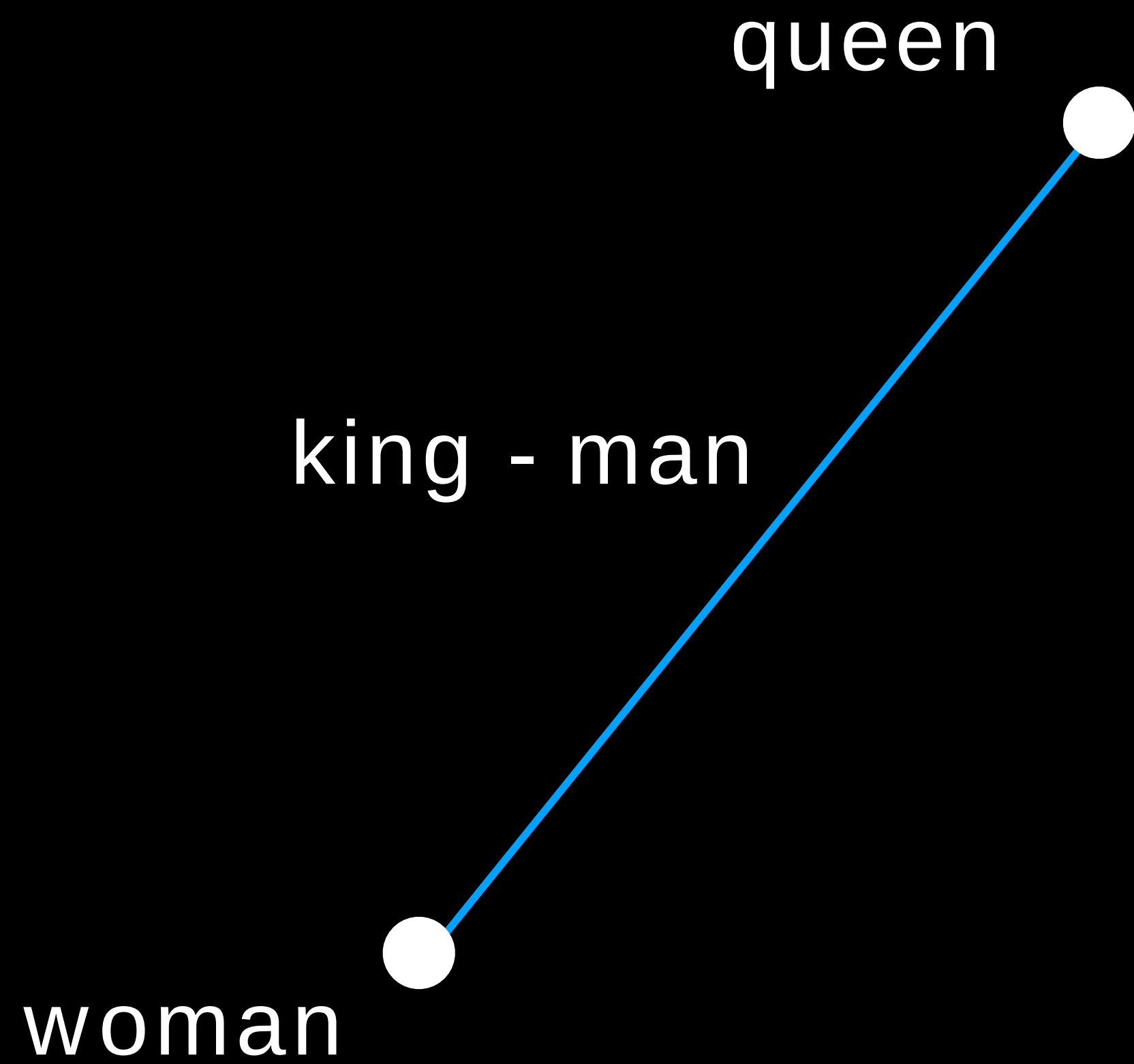
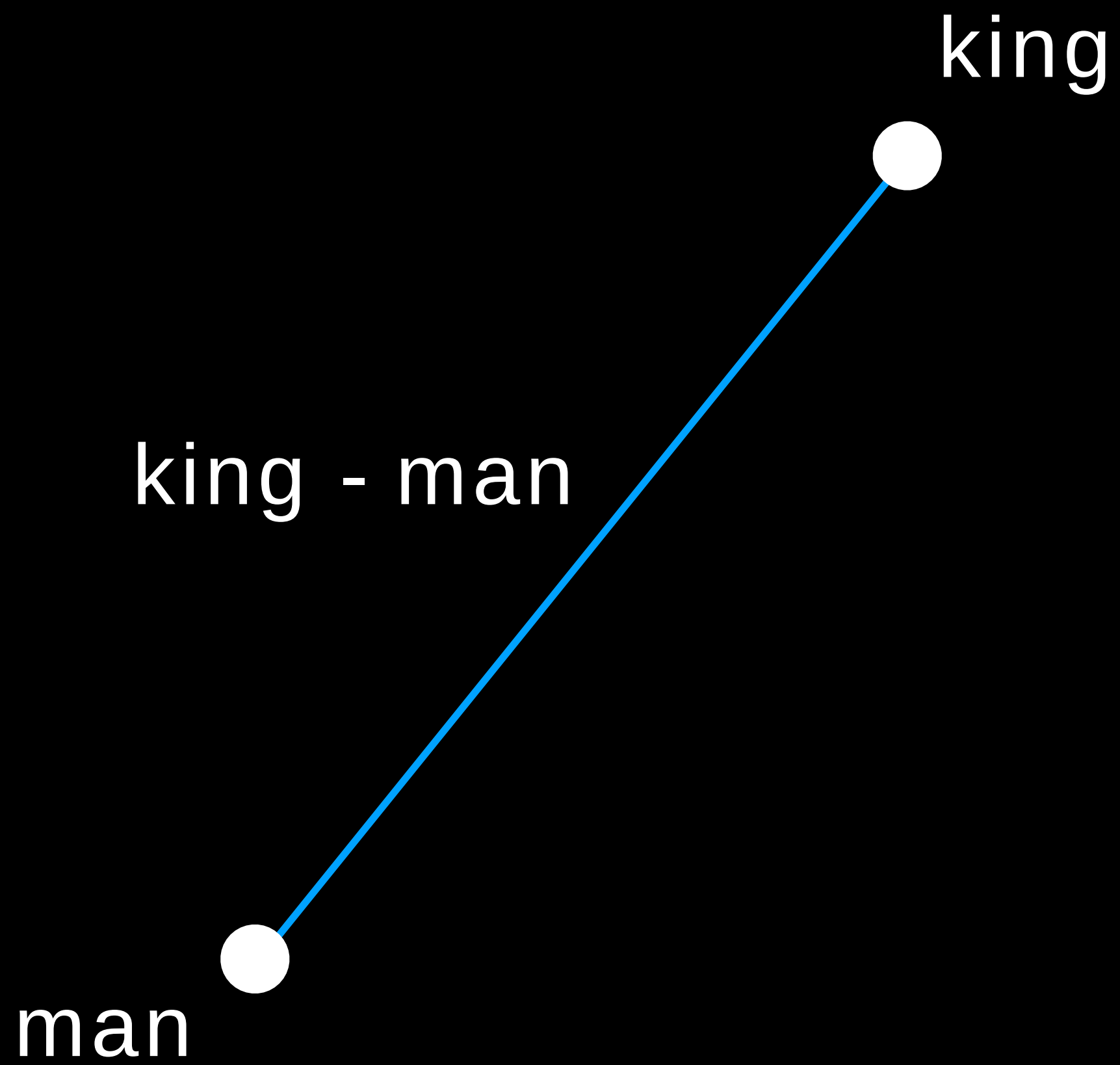
lunch

dinner

novel



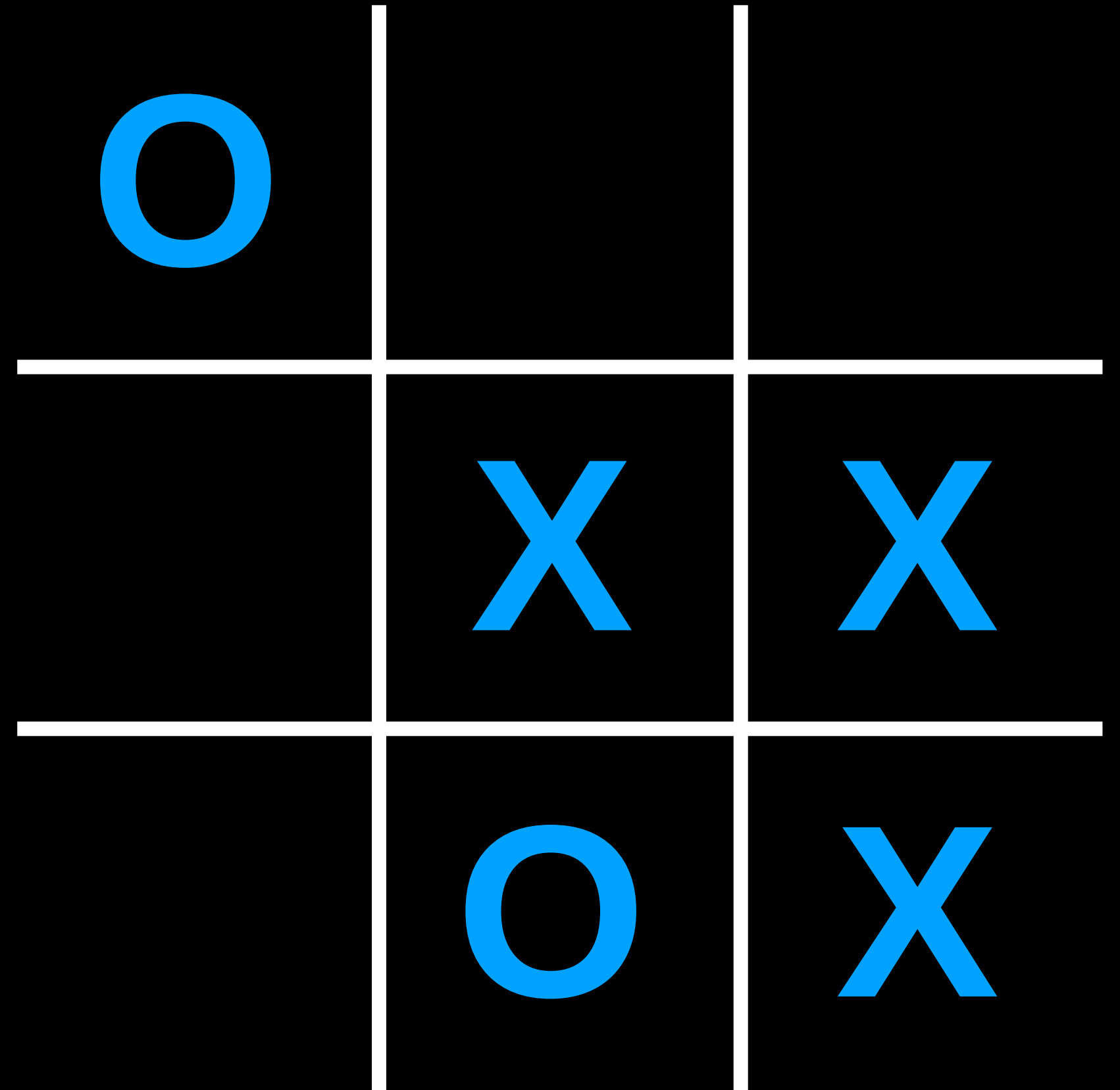




Language

Artificial Intelligence

Search

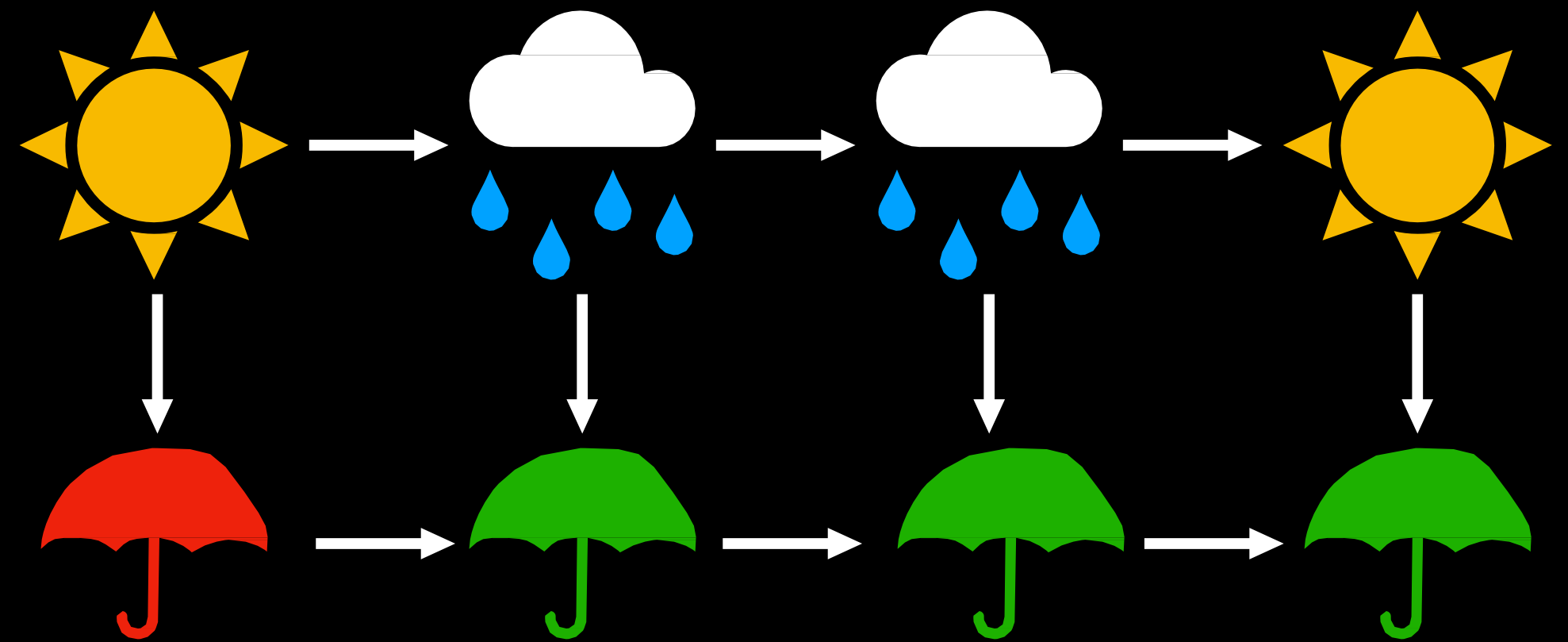


Knowledge

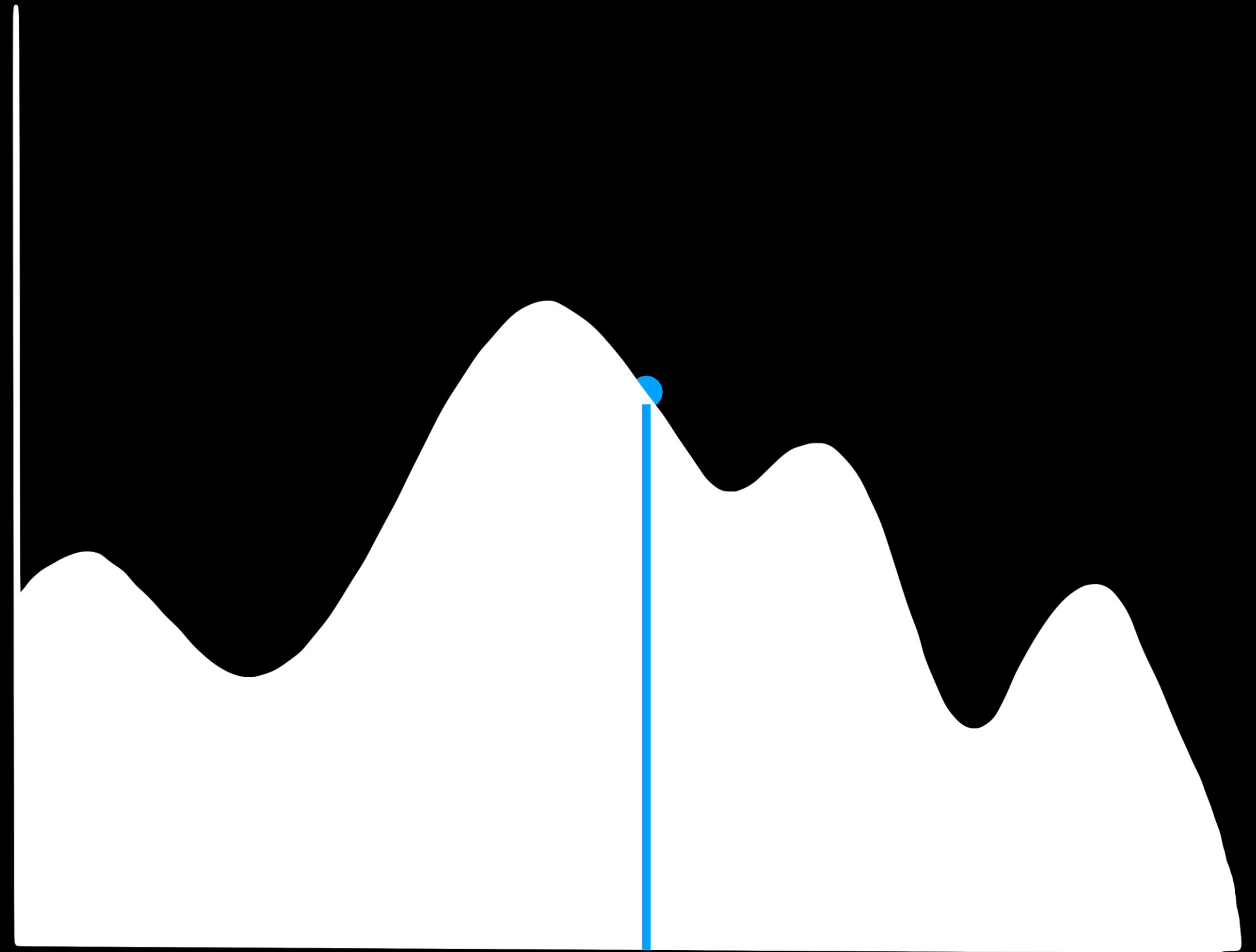
$P \rightarrow Q \quad P$

Q

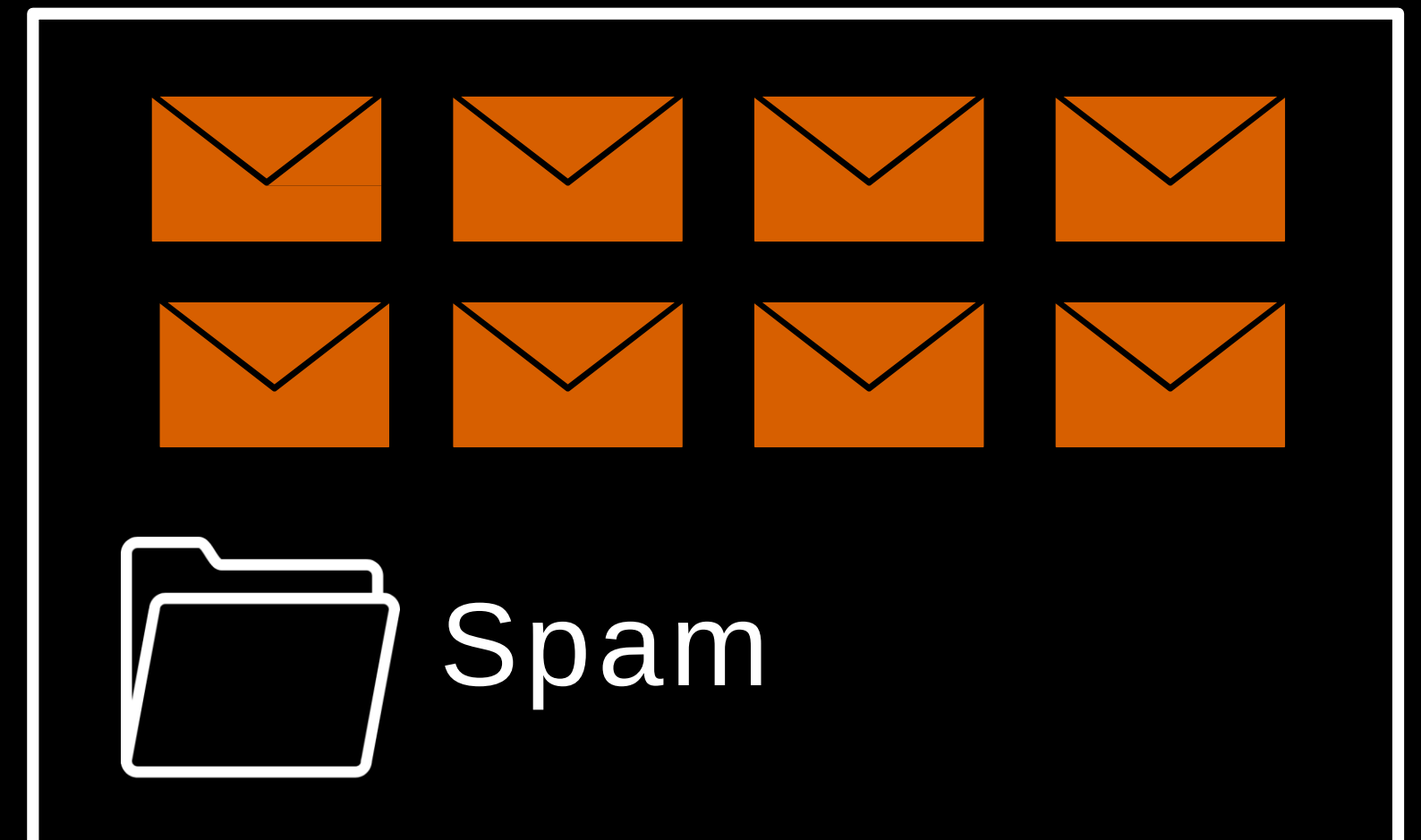
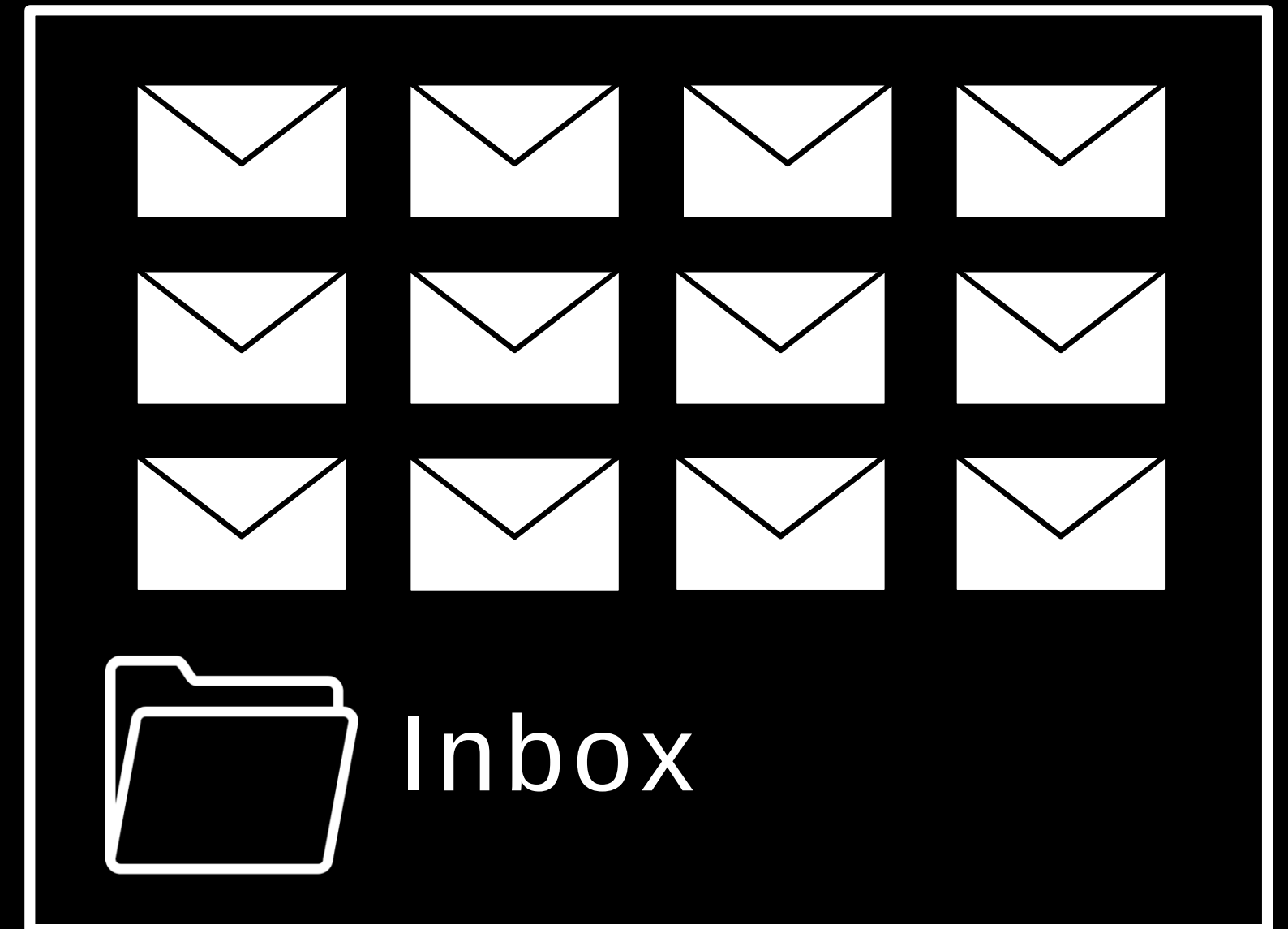
Uncertainty



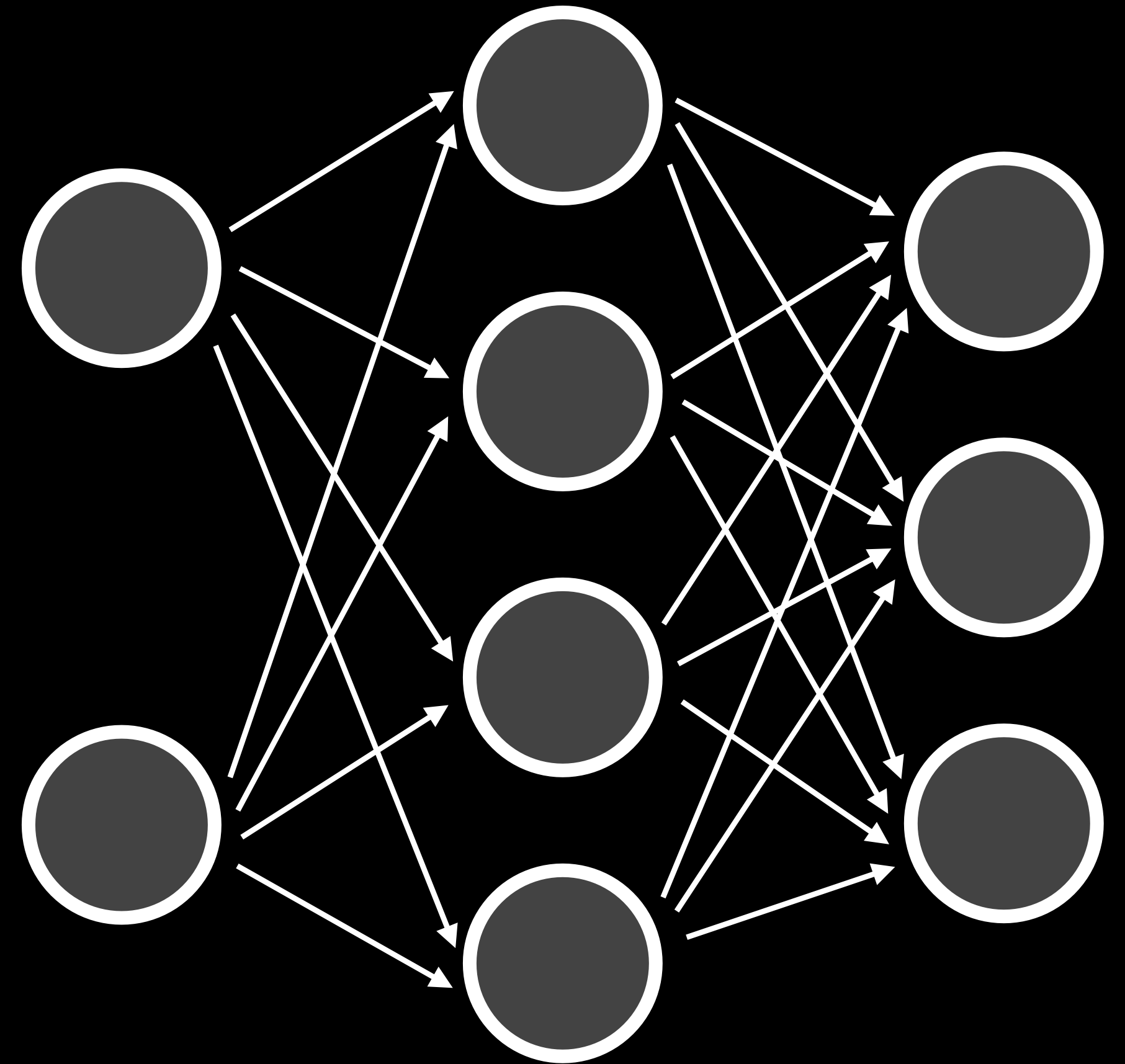
Optimization



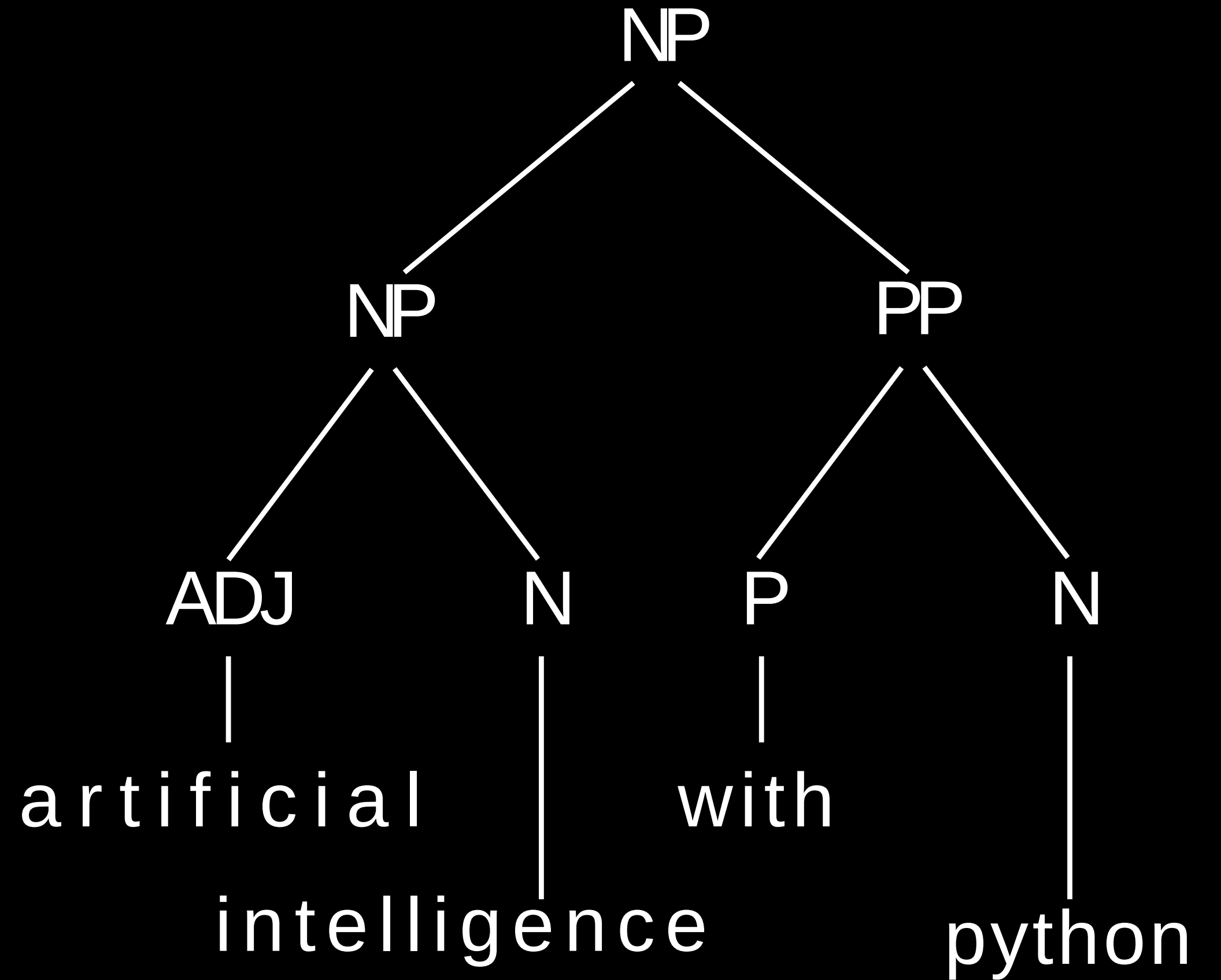
Learning



Neural Networks



Language



Introduction to Artificial Intelligence